Iowa State University
Electrical And Computer Engineering

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**EE 230**

# IR Transmitter Design

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**April 11, 2025**

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# 1 Introduction

## 1.1 Project Overview

In this design project, we were tasked with designing an Optical Signaling Device capable of generating, transmitting, receiving, distinguishing, and rectifying a signal to power an LED. This report presents our findings and solutions for generating two distinct oscillations at a specified frequency. We ensured that sufficient current was maintained in our signal to allow for transmission over a distance where it could still be received and correctly interpreted. Once we received the signal, we filtered out unwanted frequencies and rectified the oscillation so that it could be used to activate the desired LED. Our specific design was capable of distinguishing between two signals at ranges of up to 20+ feet.

## 1.2 Design Specification

- Two-button remote transmitter sending optical signals via infrared LED
- A receiver that lights either a red or green LED based on which button is pressed
- There is no illumination when no buttons are pressed
- Operational distance of up to 2+ feet between transmitter and detector
- Power supply requirements: up to three DC power supplies

## 1.3 High-level Overview

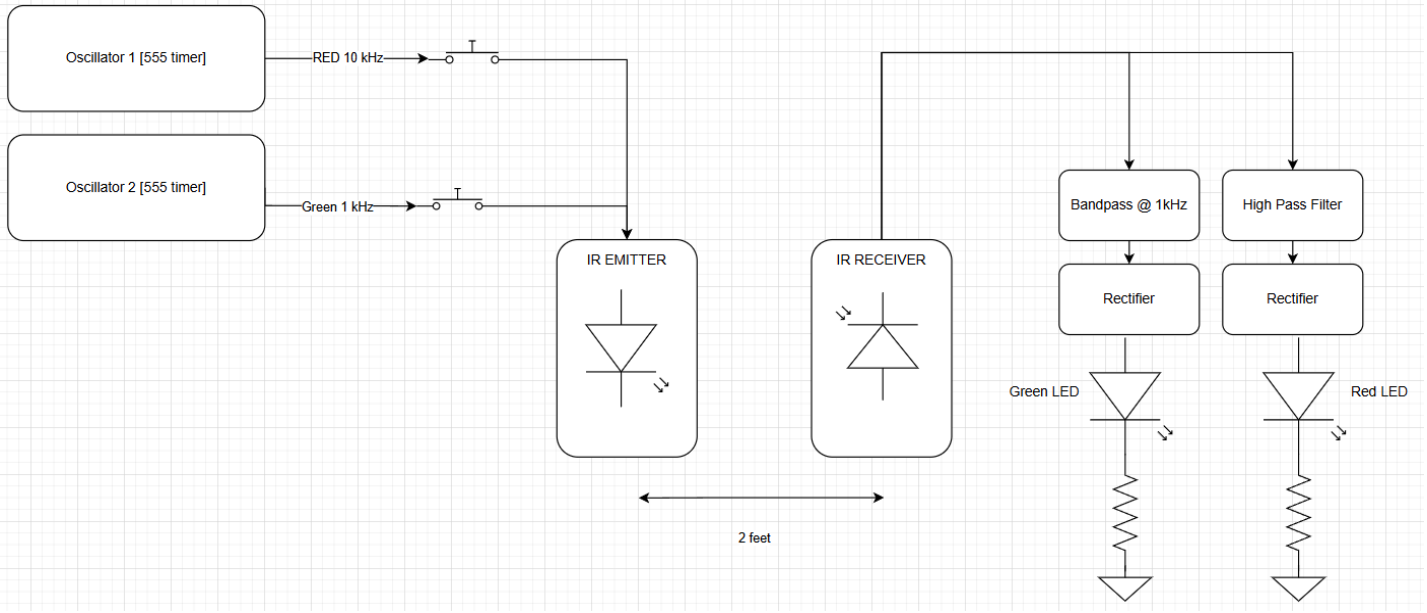
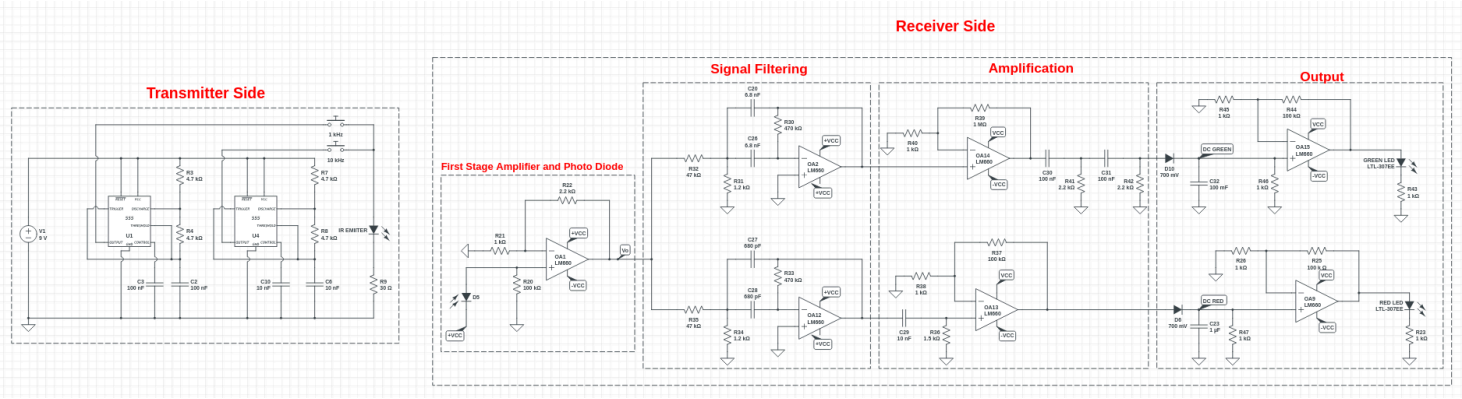



***Figure 1**: Overview of circuit functionality/flow diagram*

As mentioned in the Introduction, our primary task was to transmit either a 1 kHz or a 10 kHz signal over a specified distance. Our goal was to achieve at least two feet of transmission range while still being able to receive and distinguish between the two frequencies by filtering out the other one. When a 1 kHz signal is transmitted, we expect the green LED to turn on while the red LED remains off. Conversely, when a 10 kHz signal is transmitted, the red LED is expected to turn on while the green LED remains off; if no signal was transmitted, it was expected that neither LED would be on.



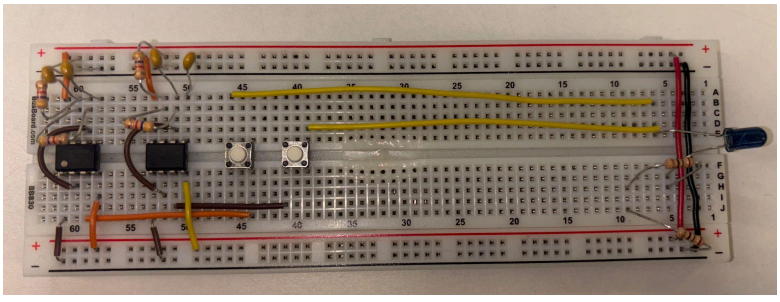



***Figure 2**: Transmitter Circuit*

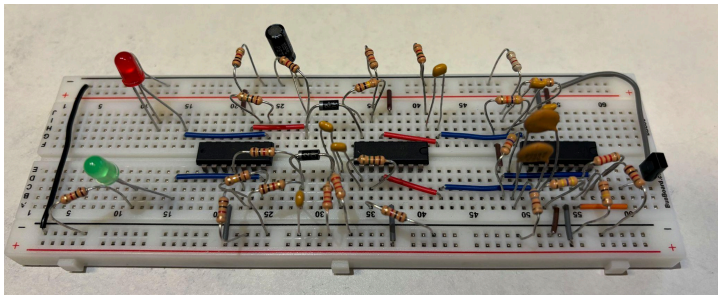


***Figure 3**: Transmitter circuit receiving circuit*

# 2 Transmitter

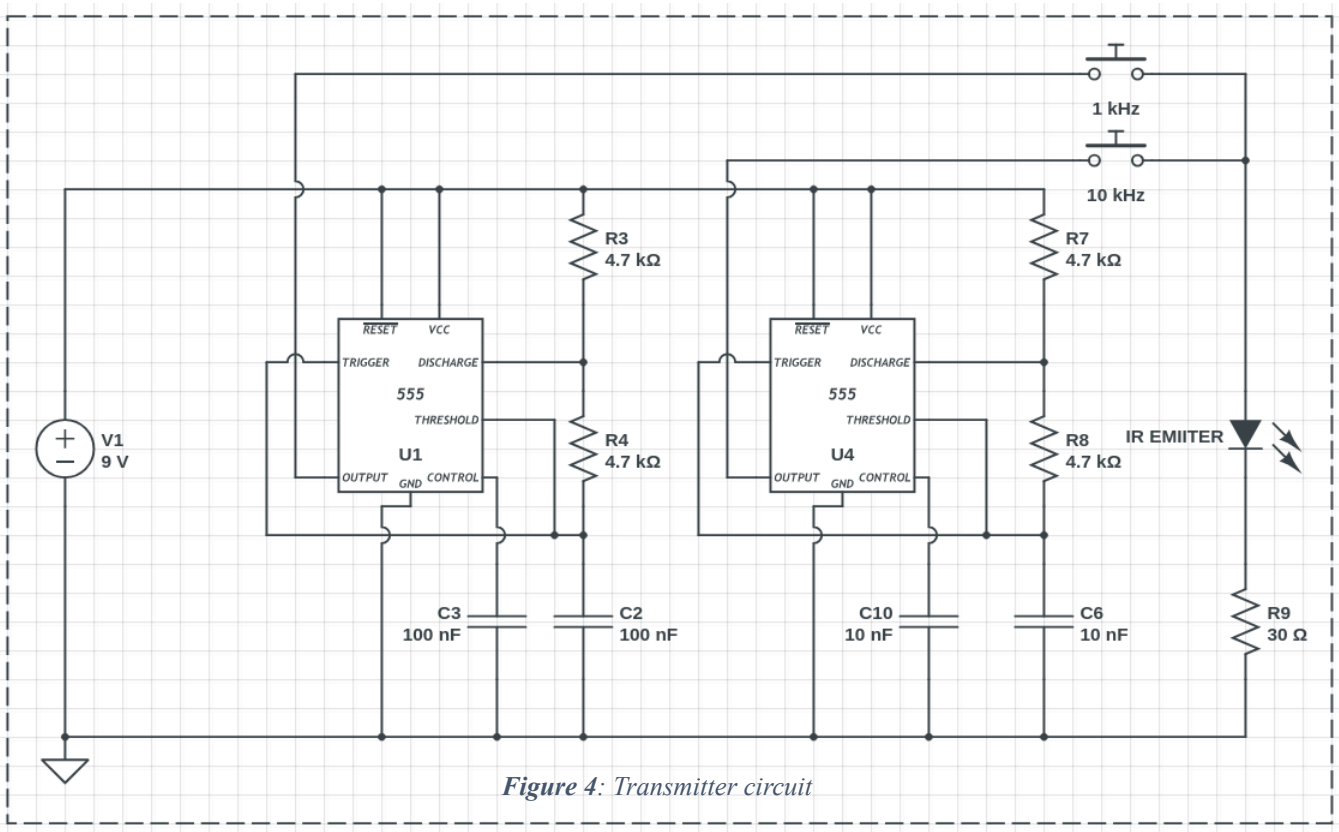



*Figure 4: Transmitter circuit*

## 2.1 Creating an Oscillator

When designing our output waveforms, we chose to go with an NE555 timer instead of an op-amp oscillator circuit such as the Wien Bridge. The primary advantage of the NE555 timer is found in the NPN BJT inside the timer (Figure 6). This circumvents the need for a current-amplifying BJT in the transmitter circuit. The max current output of the NE555 timer is 200mA, which is precisely the desired amount for our LED at 50% duty cycle. The main drawback of the NE555 timer method is that the output waveform will be a square wave instead of a sinusoidal wave. Our filtering on the receiving side mainly handles this but could potentially generate harmonics that interfere with our filtering.

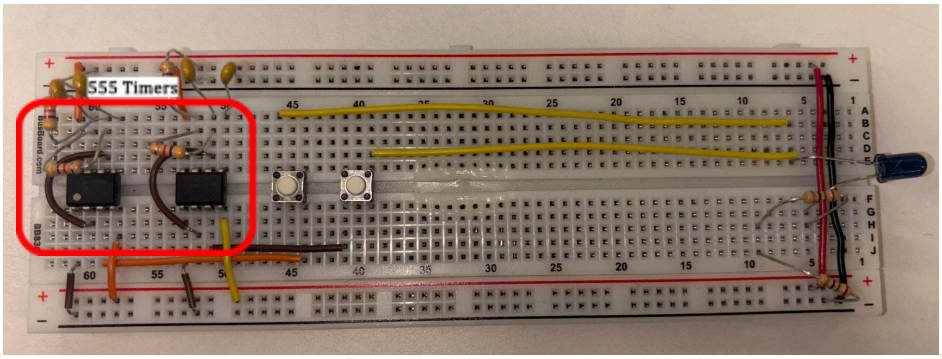



*Figure 5: 555 timer circuit picture*

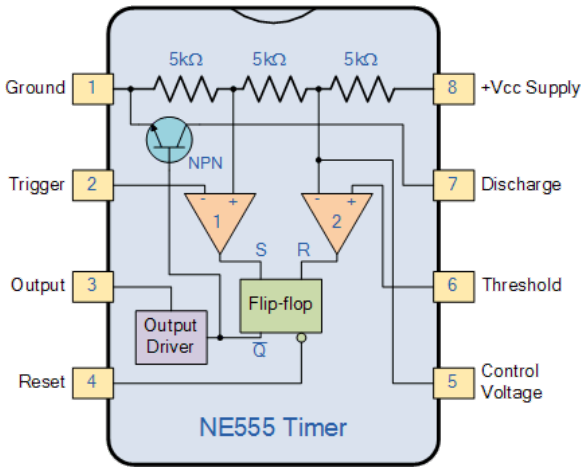



***Figure 6**: 555 timer diagram*

The inspiration for this oscillator came from Ben Eater's explanation of the Astable 555 timer, in which we took that architecture of a design. The Astable 555 timer can generate a stable oscillation with a relatively simple design; the design utilizes a comparator circuit tied to an SR latch's Set and Reset lines. With the constant charge of the output Capacitor and the discharge of the NPN BJT inside the 555, it can constantly switch between Set and Reset, which is quite beautiful since this allows us to set our own desired frequency and duty cycle of an oscillation and makes maintaining that oscillation stable. Compared to utilizing a Wien Bridge Oscillator that would require us to hold Barkhausen criteria, which, given our tolerance-based components in our lab, is hard to generate and maintain that oscillation on a breadboard without constant debugging and fixing. Thus, the idea of utilizing a 555 timer was apparent.

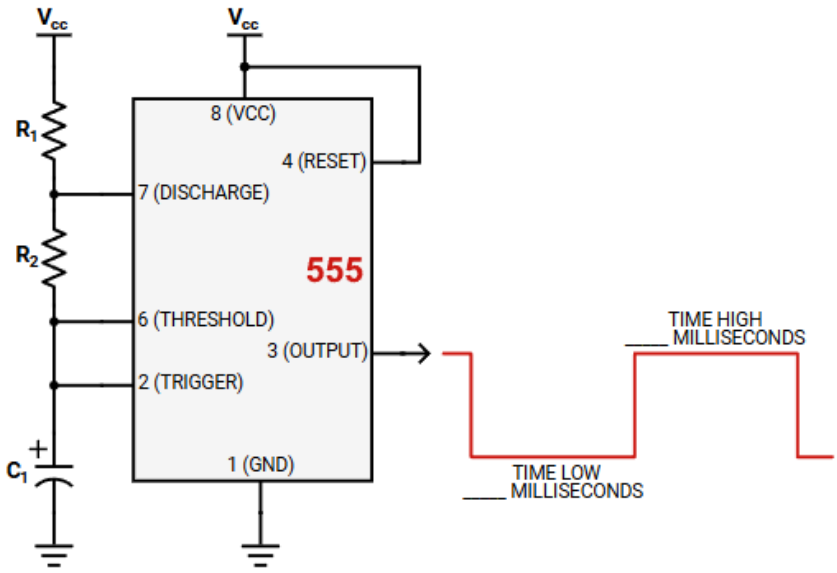



***Figure 7**: Astable 555 timer circuit*

When it came to generating our desired frequencies and duty cycle, we utilized the NE555 timer's datasheet, which specified the following equations.

$$T_h = 0.693(R_1 + R_2)C_1$$

$$T_l = 0.693R_2C_1$$

$$f = \frac{1.44}{(R_1 + 2R_2)C_1}$$

The most important aspect of the calculation was getting to that 1 kHz and 10 kHz frequency. We found the following values of our given lab kit to work well:

$$f_{10kHz} = \frac{1.44}{(R_1 + 2R_2)C_1} = \frac{1.44}{(4{,}700 + 2(4700))(10 \cdot 10^{-9})} \approx 10,212.76\ Hz$$

$$f_{1kHz} = \frac{1.44}{(R_1 + 2R_2)C_1} = \frac{1.44}{(4{,}700 + 2(4700))(100 \cdot 10^{-9})} \approx 1,021.276\ Hz$$

$$T_{high\ @\ 10kHz} = 0.693(4,700 + 4,700)(10 \cdot 10^{-9}) \approx 65.142\ \mu s$$

$$T_{low\ @\ 10kHz} = 0.693(4700)(10 \cdot 10^{-9}) \approx 32.571\ \mu s$$

$$T_{high\ @\ 1kHz} = 0.693(4,700 + 4,700)(100 \cdot 10^{-9}) \approx 651.42\ \mu s$$

$$T_{low\ @\ 1kHz} = 0.693(4700)(100 \cdot 10^{-9}) \approx 325.71\ \mu s$$

$$Duty\ Cycle\ \% = \frac{T_{high\ @\ 1kHz}}{T_{high\ @\ 1kHz} + T_{low\ @\ 1kHz}} \cdot 100 = \frac{651.42\ \mu s}{651.42\ \mu s + 32571\ \mu s} \cdot 100 \approx 66.67\%$$

## 2.3 Powering the Infrared LED

We use an LED in the infrared emission spectrum for the oscillator signals to be transmitted wirelessly. We are using this wavelength as it is low-power, invisible to the eye, and relatively easy to implement. We are using the TSAL6100 infrared LED for this project, as it was available to us in our lab kit.

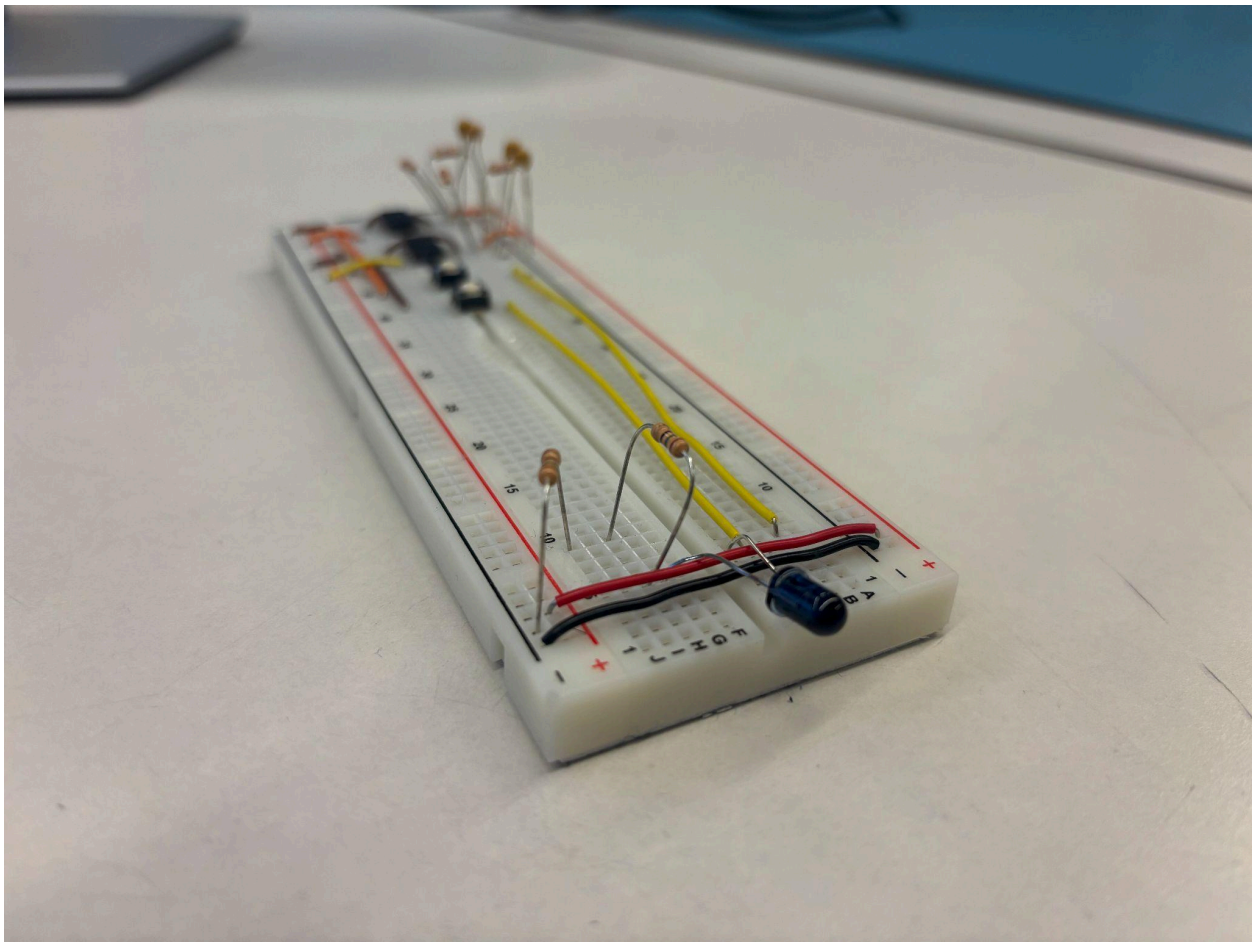


***Figure 8**: Picture of transmitter*

From the IR LED datasheet, the maximum current for this LED is around 200mA continuously, but for the given for our design, we push that a bit at 110mA average with our choice of current limiting resistors. This is still within range for our use case as the LED is not on for long periods and is not expected to have a long lifespan. We desired this extra current to facilitate a longer-range signal transmission.

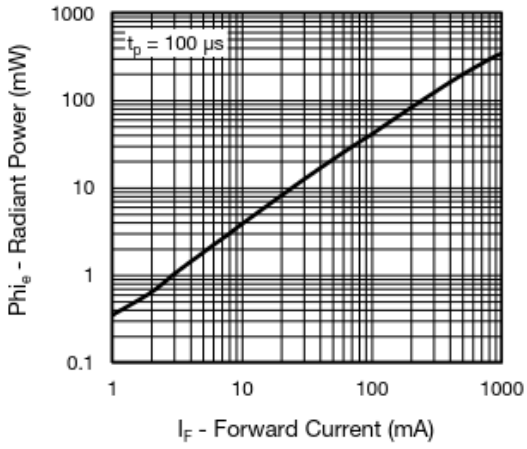



***Figure 9**: Plot of TSAL6100 radiant power vs forward current*

The radiant power increases linearly with the current, so we expect a 10% increase in power output (Figure X). With an output from the NE555 timers at ~7.8V and a ~1.3V forward bias voltage drop across the LED, we can expect a remaining voltage of around 6.5V. With our timer at around a 50% duty cycle, we desire a current of about 200mA peak and 100 mA average.

$$(200mA) \cdot 0.50 = 100mA$$

This can be achieved using 30 Ω of resistance. The estimated power of a single 30Ω resistor at this current (100mA) would be 300mW.

$$(0.1A)^2 \cdot 30\Omega = 300mW$$

The resistors we used were only rated for 250mW, so we used two 15 Ω resistors in series to half the power dissipated by each.

# 3 Receiver Circuit

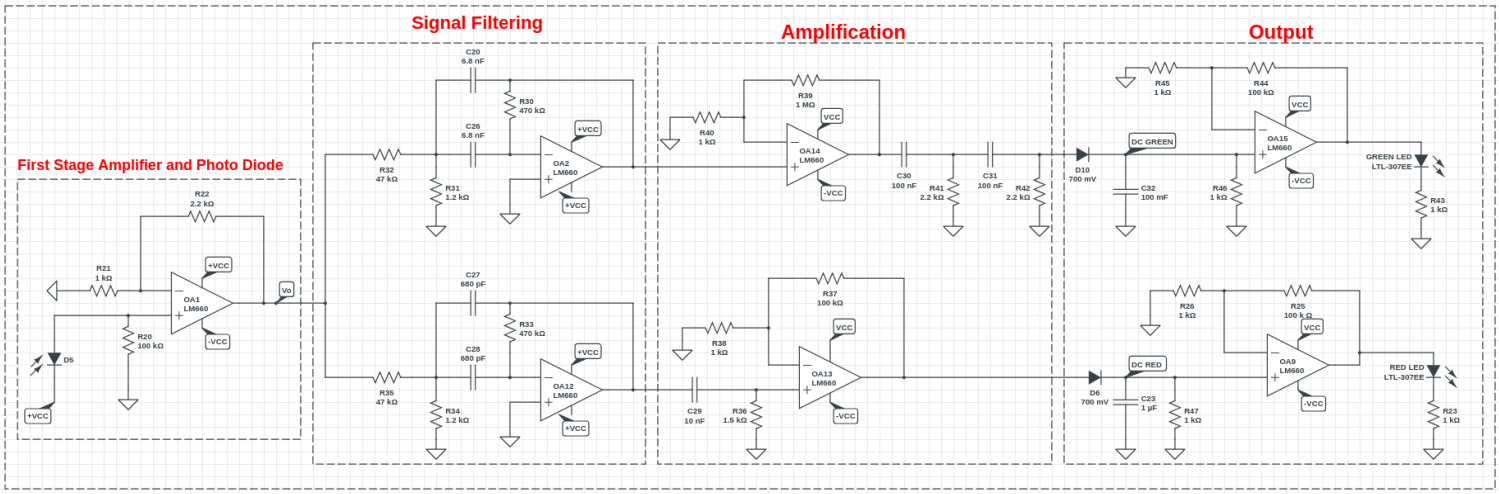



*Figure 10*: *Top-level receiver circuit*

## 3.1 First Stage Amplifier and photodiode

On a secondary breadboard, we begin our infrared receiver circuit. The first stage of this circuit is responsible for amplifying the signals we receive from the transmitter. We do this through a Transimpedance Amplifier (TIA), which converts an input current to an amplified output voltage.

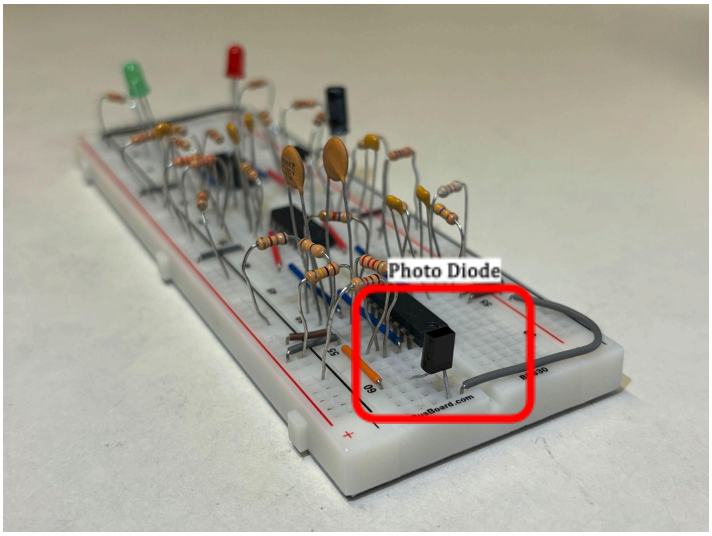



*Figure 11*: *Receiver circuit picture (photodiode)*

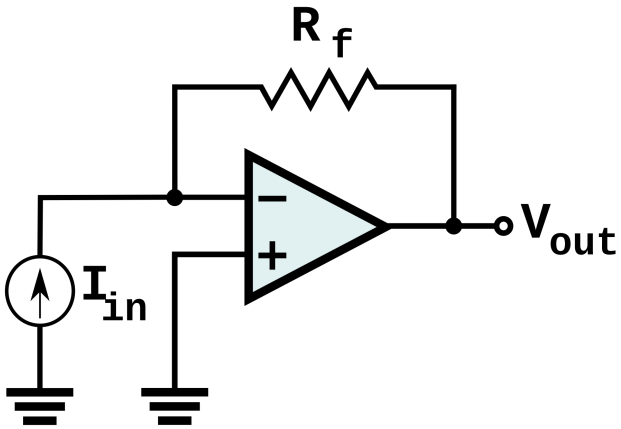



*Figure 12*: *Example Transimpedance Amplifier circuit*

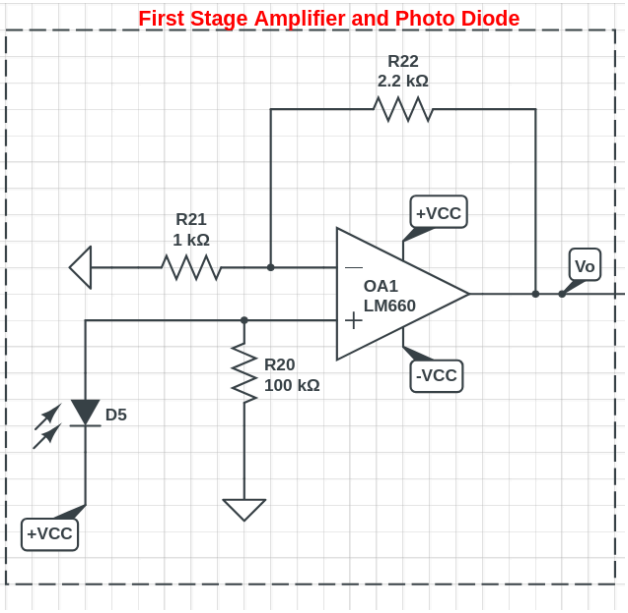



***Figure 13**: Project trans-impedance amplifier circuit*

In our case, the current source is a reverse-biased photodiode that responds to the infrared emissions from the transmitter circuit. We chose to use a non-inverting version of a TIA as it fixed some oscillation issues believed to be caused by the non-ideal op-amp.
This stage aims to get a "line-level" signal that could be filtered easily in later stages. When the infrared diode is placed in line of sight at 4 feet, the photodiode sees ~48µA. This current is highly affected by the facing angle of the LED, with the current becoming almost unnoticeable at angles greater than 20 degrees. The distance from the receiver also changes the current by the inverse square law (twice the distance is a fourth of the intensity). We scaled the desired functional ~48 µA to ~16.4 mV with this TIA. This number can be scaled much higher if desired, but it worked for our design. It may be possible to reduce the total components in the receiver circuit by increasing the output gain of this op-amp and removing an amplifier later.

In the scope below, we can see that we are reading a 4.8mV reading on the cathode end of the photodiode, This voltage is coming from the 10 kHz transmitter approximately 4 feet away.

After passing through the trans-impedance amplifier, we can see our input signal has now been scaled up slightly over 4 times the input value. Looking back, we should have inserted a higher feedback resistor to increase the gain further to reduce the number of operational amplifiers used in this circuit, especially since we would not have had any issue with the Gain Bandwidth product at 10 kHz.

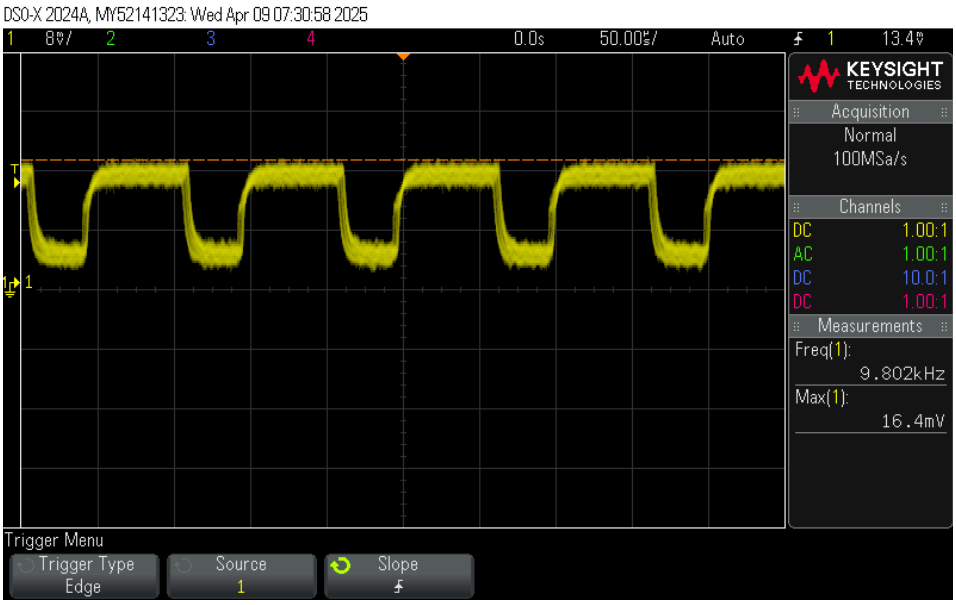



***Figure 14**: Scope trace post-trans-impedance amplifier*

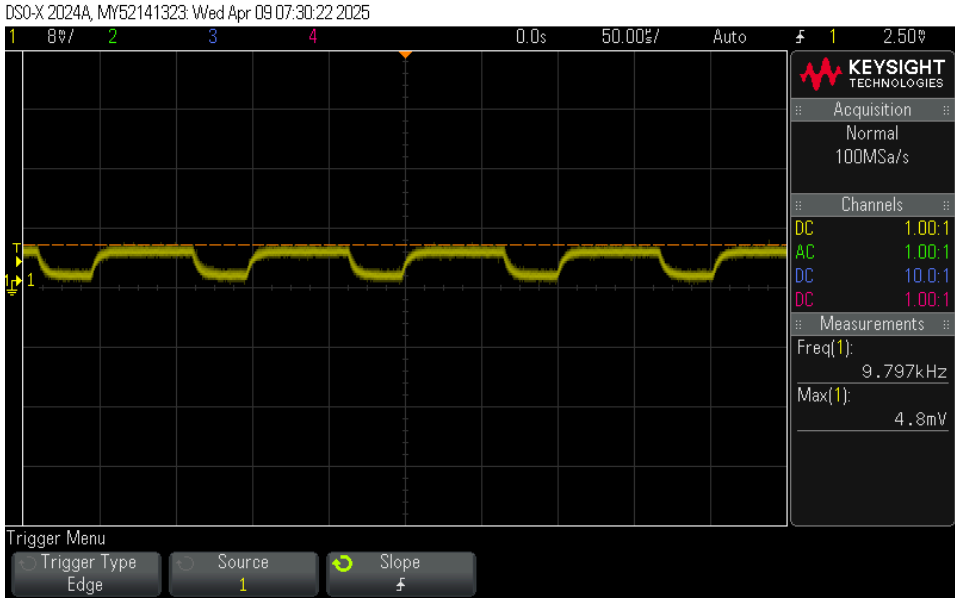



***Figure 15**: Scope trace pre-trans-impedance amplifier*

## 3.2 Signal Filtering & Amplification

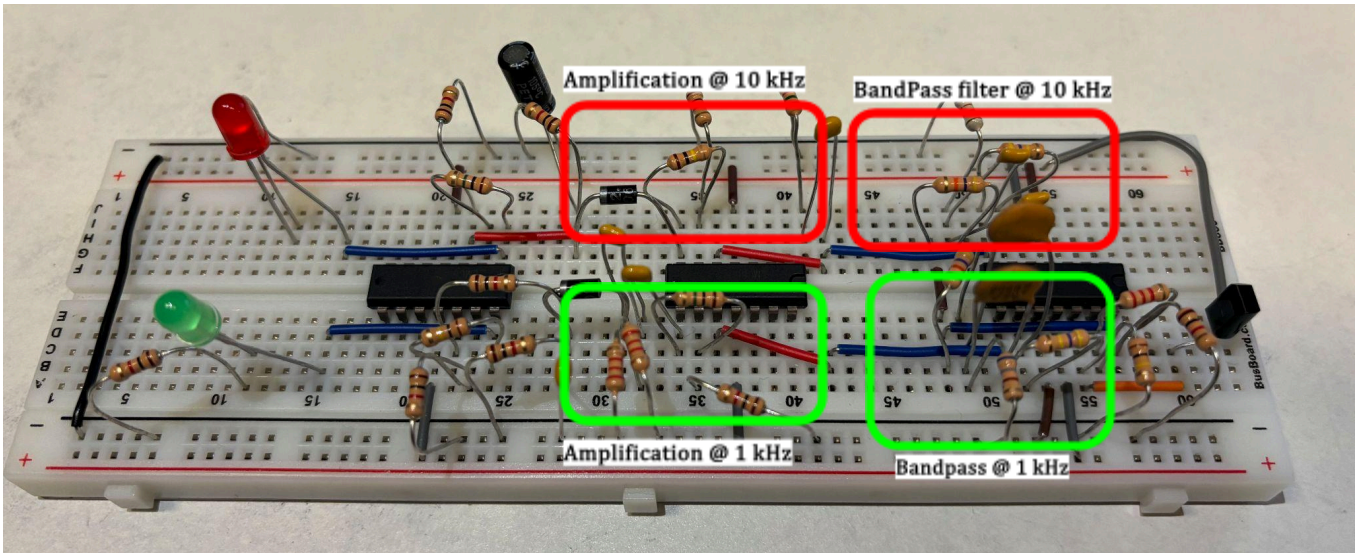



***Figure 16**: Filter and amplification circuit picture*

To differentiate between the 1 kHz and 10 kHz signals, the amplified signal is passed into two Deliyannis-Friend second-order bandpass filters. A bandpass filter was chosen over a low-pass and high-pass for each respective signal because it offered more precise filtering. This way, we could filter out any unwanted signals we didn't plan for, including upper harmonics from the square waves we used for transmitting. We chose the Deliyannis-Friend filter because it offered a large Quality factor and was simple to solve. We used Gary Tuttle's website as a reference when designing and building our filters. The first filter is tuned for 1 kHz, and the second is tuned to 10 kHz. To account for variation in component values on the desired operation, an AC frequency response analysis tool called Loopinator was used. Using the results of this tool, we tweaked values to move the characteristic frequency exactly on 1kHz and 10kHz.

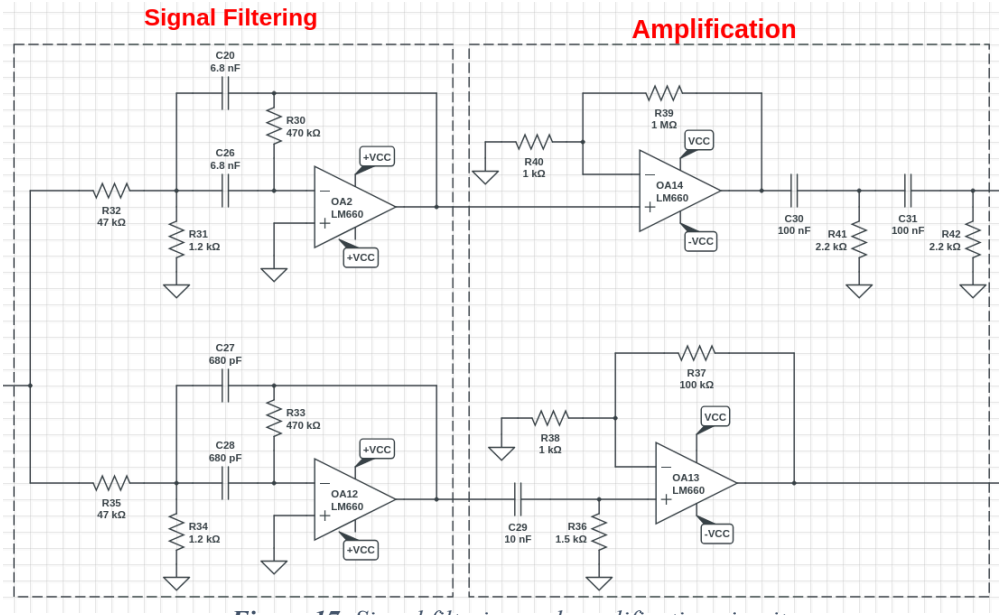



***Figure 17**: Signal filtering and amplification circuits*

1. $\omega_o = 2\pi f_o = 6280$ rad/s.
2. $\omega_o / Q_P = \Delta\omega$ (bandwidth) $\rightarrow Q_P = f_o / \Delta f = 10.$
3. Choose $C_1 = C_2 = 6.8$ nF.
4. $Q_P = \frac{1}{2}\omega_o R_2 C \quad \rightarrow \quad R_2 = \frac{2Q_P}{\omega_o C} = 468\,\text{k}\Omega$
5. $G_o = \frac{R_2}{2R_1} \quad \rightarrow \quad R_1 = \frac{R_2}{2G_o} = 46.8\,\text{k}\Omega$
6. $\omega_o^2 = \frac{1 + \frac{R_1}{R_3}}{R_1 R_2 C^2} \quad \rightarrow \quad R_3 = \frac{R_1}{\omega_o^2 R_1 R_2 C^2 - 1} = 1.20\,\text{k}\Omega$

***Figure 18**: Daliyannis-Friend filter equations from Gary Tuttle's website [4]*

The example problem on Gary Tuttle's website happened to work perfectly for our design, so we used the same values. In order to solve for the 10kHz band pass filter we chose to decrease both capacitor values by a factor of 10 (680pF). This lets us use all of the same resistor values as well as the other assumption.

Immediately succeeding the bandpass filtering, a second-stage amplifier and a high-pass filter are used. This stage gives a gain of ~100 for the 10kz signal and ~1000 for the 1kHz signal. We had to use a lower value for the 10kHz signal to stay beneath the Gain-Bandwidth product of 1.4MHZ for the LM660 op-amp we used. The high-pass filters nullify any DC offset present after the first stage and any low-frequency mains interference (60Hz).

## 3.3 LED Power Circuit

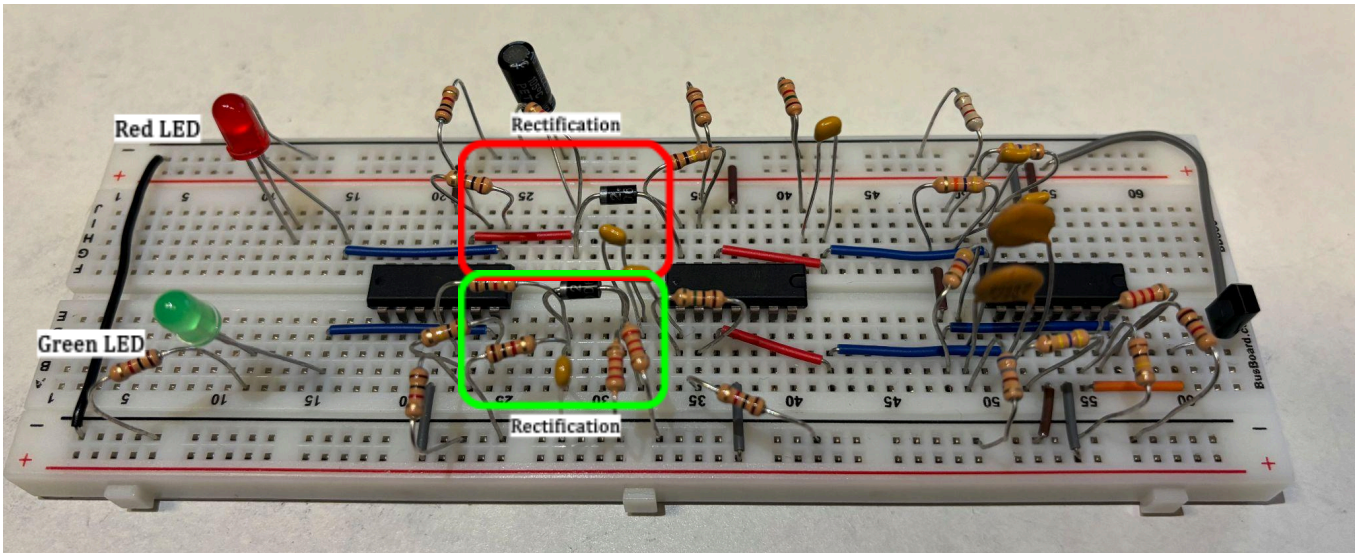



***Figure 19**: Rectification circuit picture*

The final stage of our receiver circuit includes a half-wave rectifier and a comparator feeding an LED with a current-limiting resistor.

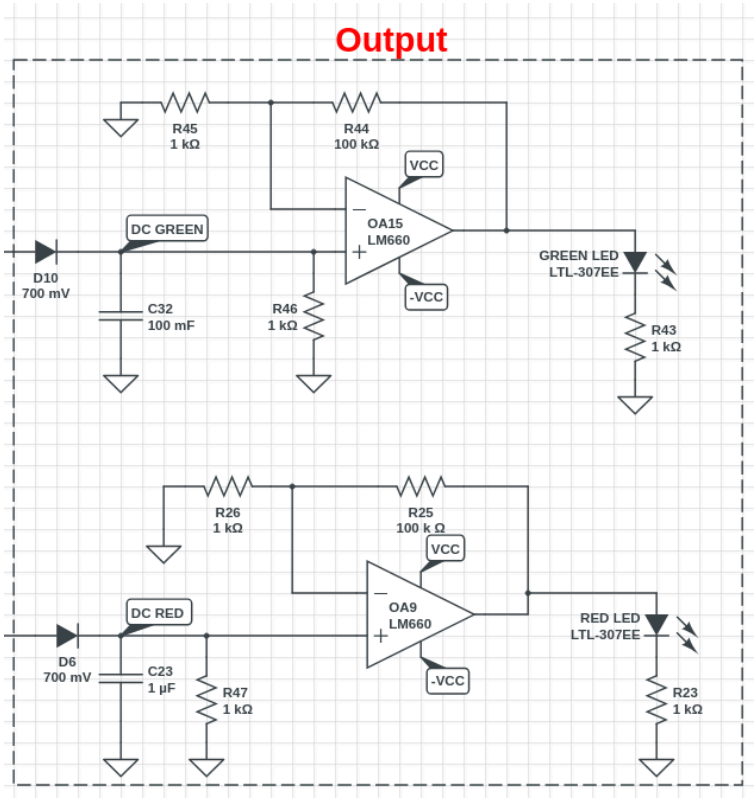



***Figure 20**: Rectification circuit diagram*

The amplified signal from the filtering circuit is passed through a half-wave rectifier, which is then amplified by a gain of about ~100, which will act as a comparator to create a solid DC signal at saturation. This produces a DC signal of 8V (Vcc) that is fed straight into the LED and a current-limiting resistor. The 1k Hz circuit feeds a green LED, and the 10kHz feeds a red LED, both with an observed forward voltage drop of ~2V. With a 1kΩ resistor in series, we get a current of ~6mA, enough to illuminate the LED. This 1kΩ resistor can be traded out for a lower resistance if a brighter LED output is desired.

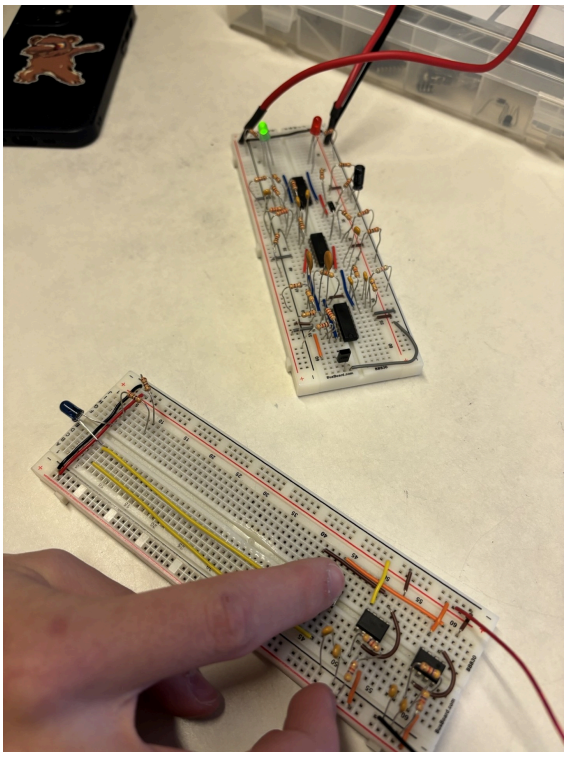


***Figure 21**: 1 kHz properly turning on Green LED*

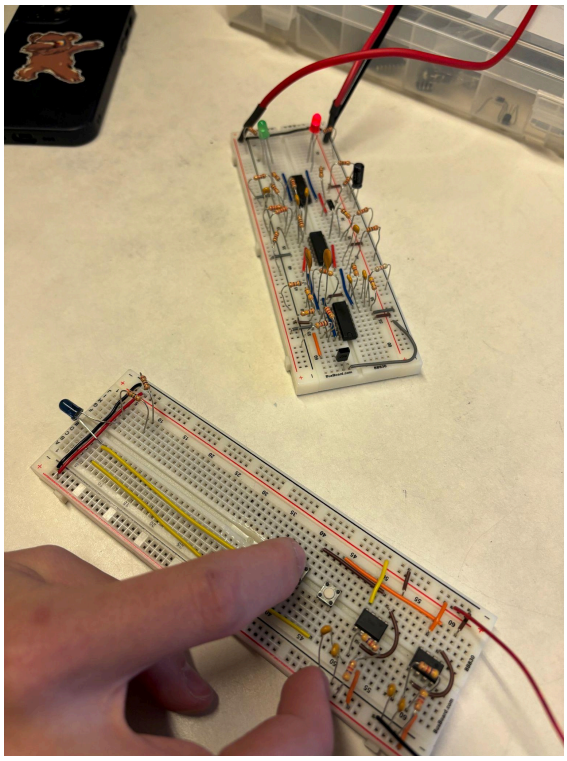


***Figure 22**: 10 kHz properly turning on Red LED*

The low current works well in this case as the output from the final op-amp is essentially digital (either 8v or 0v), and the low current will not pose any power dissipation problems.

# Simulation Results

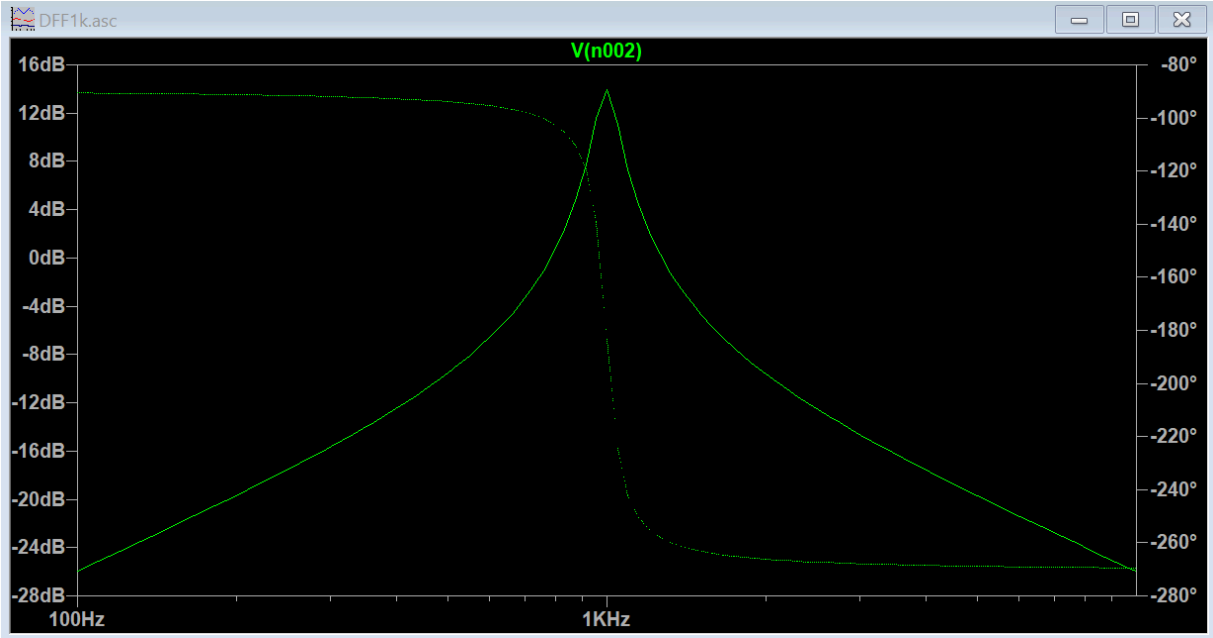



***Figure 23**: Plot of 1 kHz Deliyannis-Friend Second Order Filter*

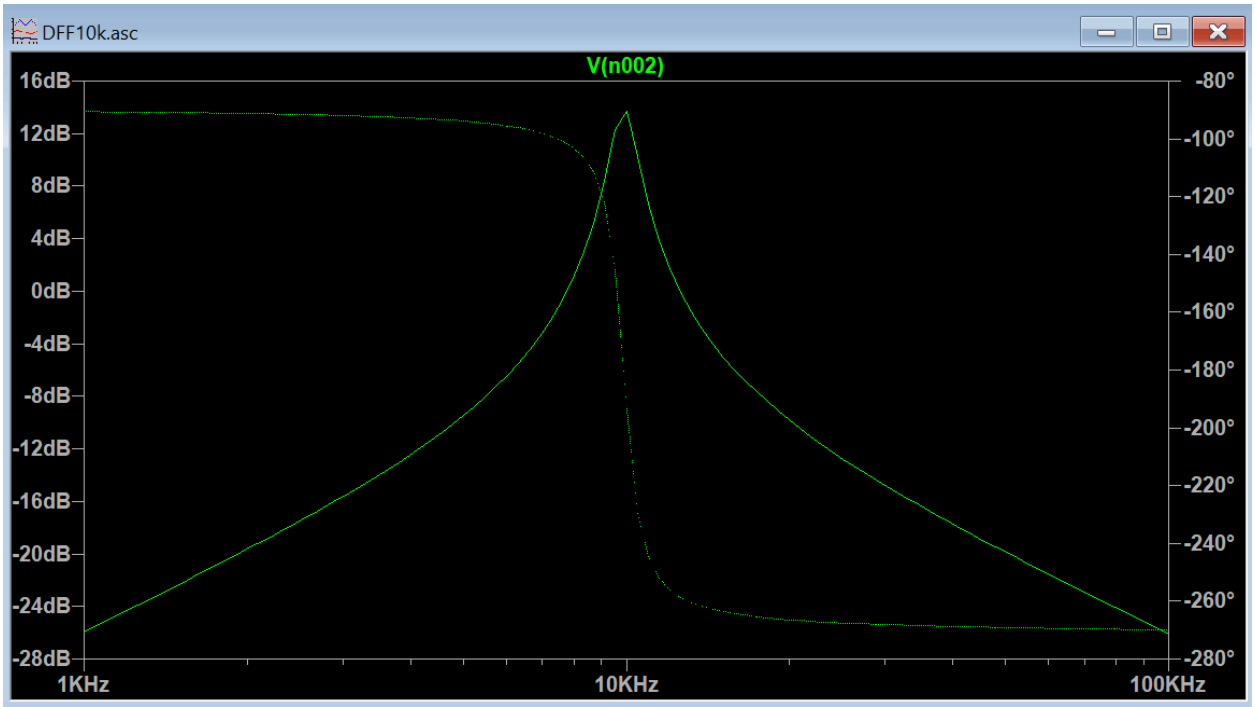



***Figure 24**: Plot of 10 kHz Deliyannis-Friend Second Order Filter*

# Plots

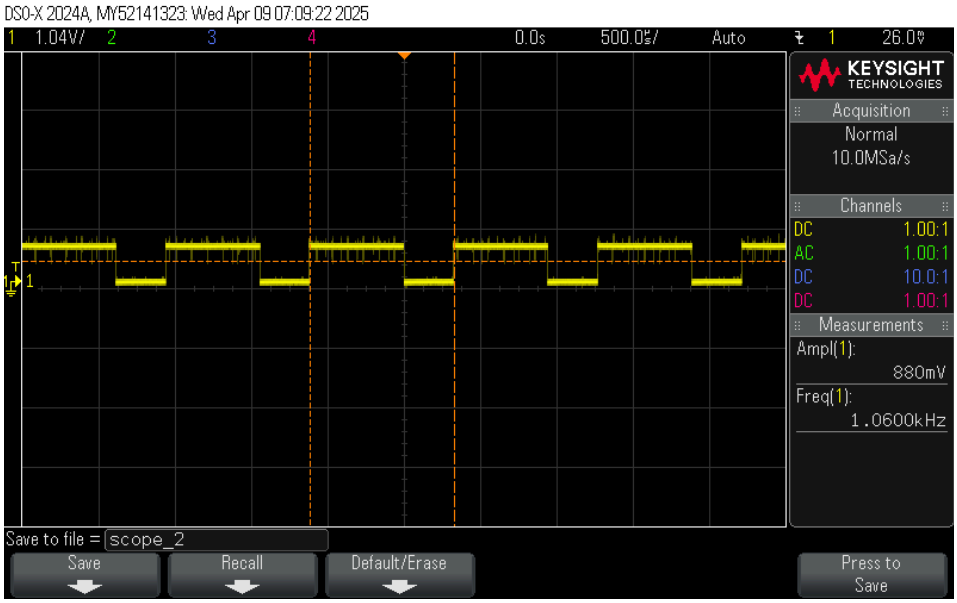



***Figure 25**: Plot of 1 kHz square wave oscillating from 555 timer*

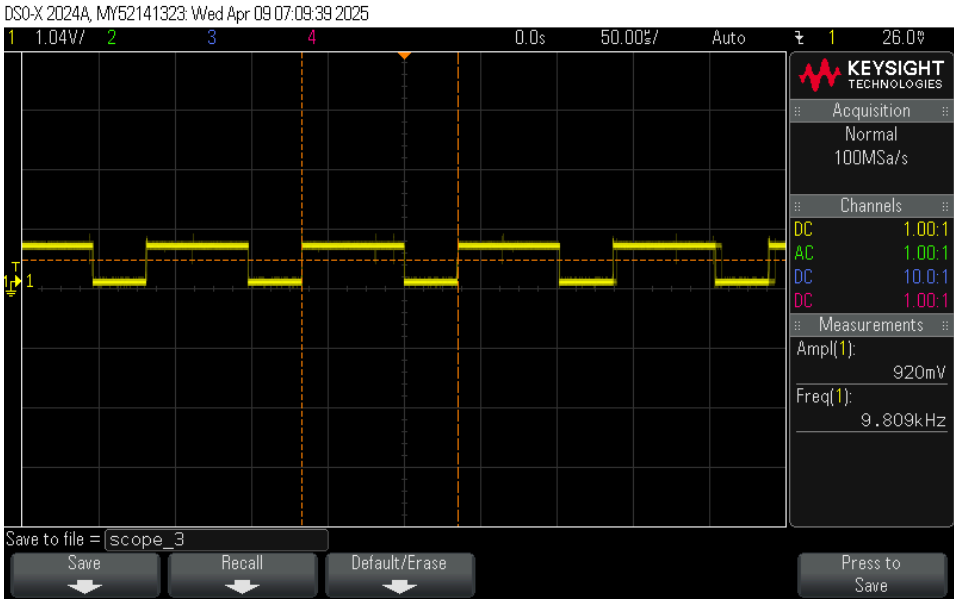



***Figure 26**: Plot of 10 kHz square wave oscillating from 555 timer*

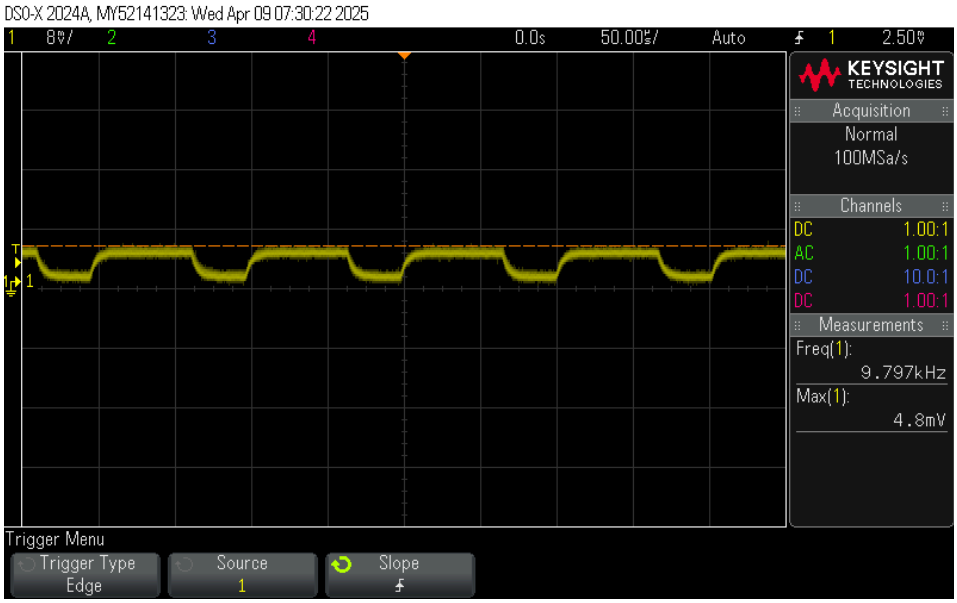



***Figure 27**: Low voltage signal received at photodiode*

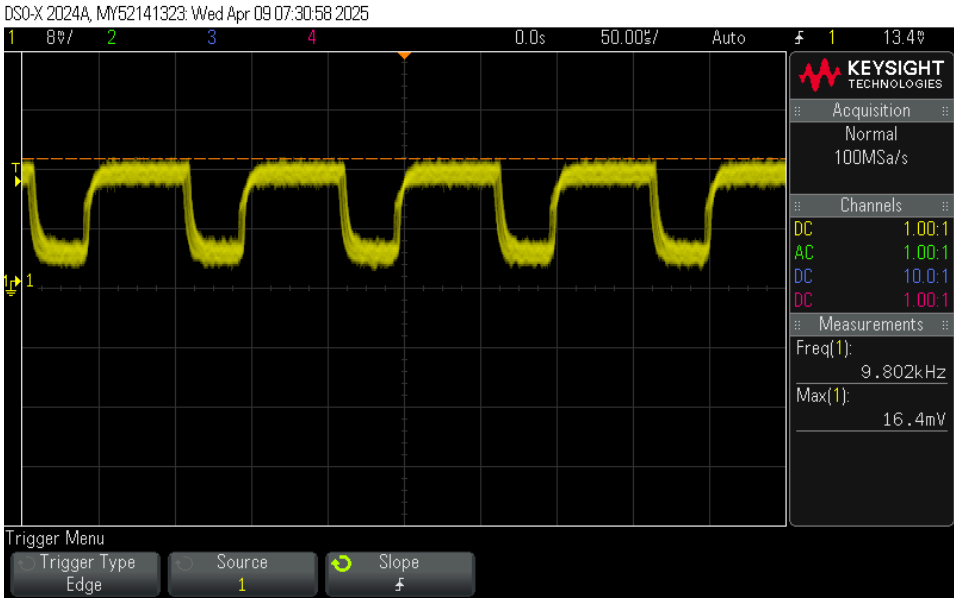



***Figure 28**: Output of trans-impedance amplifier*

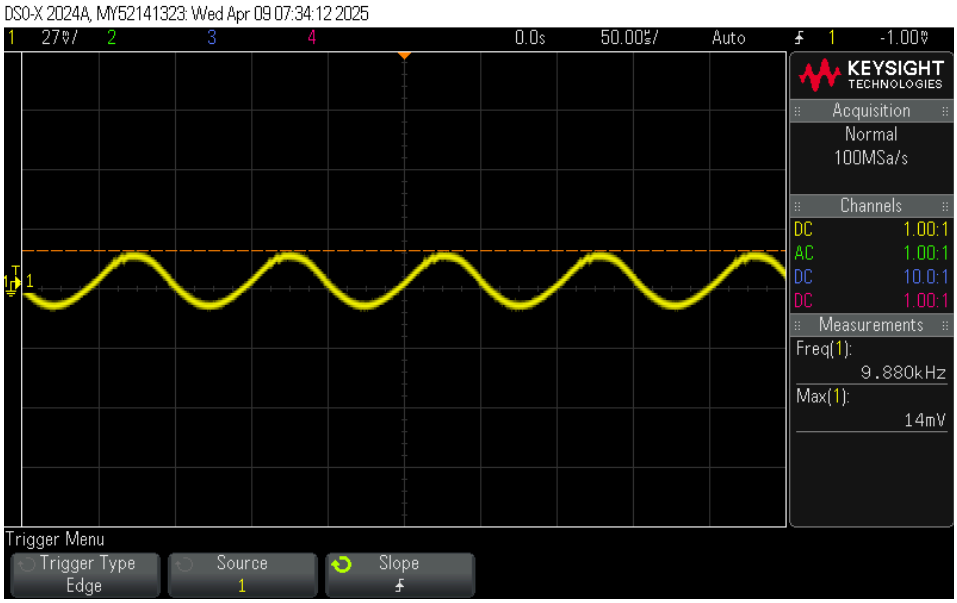



***Figure 29**: Output of 10 kHz Bandpass filter of 10 kHz input*

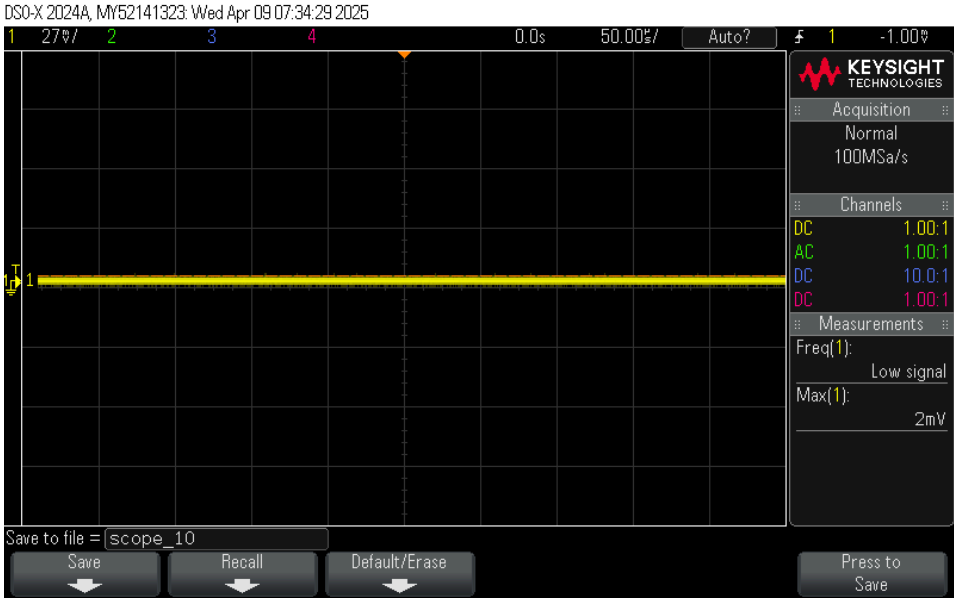



***Figure 30**: Output of 1 kHz Bandpass filter of 10 kHz input*

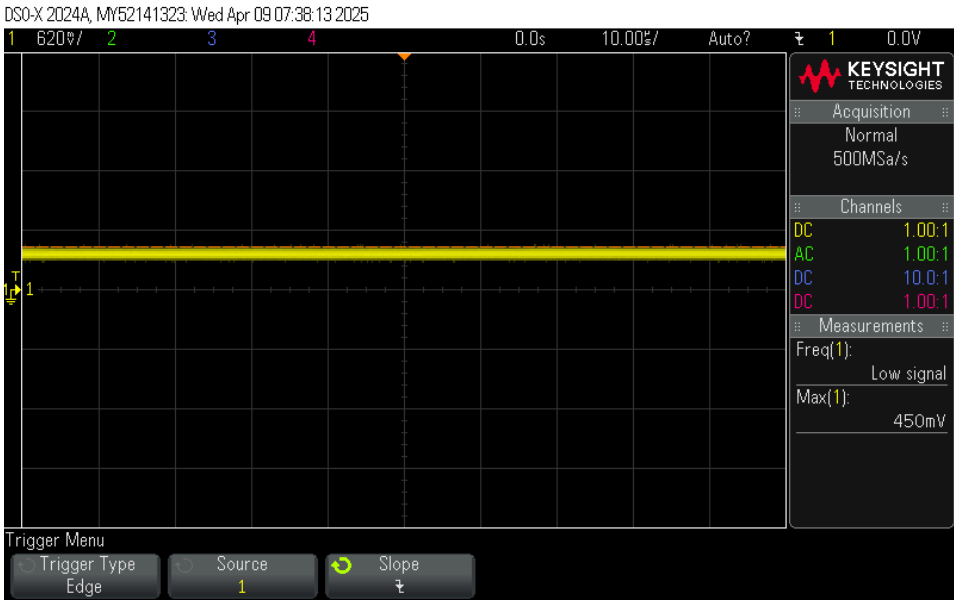



***Figure 31**: Rectified 10 kHz signal*

## Results

The final design successfully completed all design requirements and exceeded the range requirement considerably. The receiver was able to receive values from over 20 feet away which is an order of magnitude more than the requirement. The transmitter design worked perfectly with a 9V battery as a power supply, which allowed us to have a completely tether-free remote.

## Reflection

Though this design exceeds our expectations, it is not without room for improvement. At ranges of less than 1 foot we see both LEDs turn on. This is caused by our filters not fully attenuating the unwanted signal, so when the transmitter is in close proximity, the photodiode output becomes too much for the circuit to filter. To fix this, we could increase the filter order to increase the roll-off. Another possible alternative would be to increase the Q value and decrease the gain, functionally decreasing the bandwidth. Another place for improvement is optimizing the number of components used. While designing our filters, we troubleshooted erroneous signals as they appeared, for example, by adding a high-pass filter to remove a DC offset. These were not anticipated and thus were not initially accounted for. It is entirely possible that we could redesign all of the filters used into a single filter for each of the 1kHz and 10kHz signals.
In the same way, we did not need to use three different LM660 op-amps. We could have gotten away with just one or two op amps for each of the 1kHz and 10kHz signals. In this case, it would have been possible to reduce the number of LM660 modules to one instead of three.

In this project, we got to experience firsthand how to design a functional circuit from design requirements. The initial design included many errors and unnecessary components, such as a current amplifying BJT. However, through this process of trial and error, we got to experience what it takes to engineer a design from scratch. This experience has taught us about generating, transmitting, receiving, distinguishing, and rectifying a signal. As well as reinforcing our knowledge of circuit analyses with linear components and some non-linear components.

# Component Values

## Transmitter Component

| Reference | Value | Component Type | Notes |
|---|---|---|---|
| V1 | 9 V | Power Supply | Battery supply |
| U1 | 555 Timer | IC | Astable oscillator |
| U4 | 555 Timer | IC | Astable oscillator |
| R3 | 4.7 kΩ | Resistor | Based on the desired Frequency |
| R4 | 4.7 kΩ | Resistor | Based on the desired Frequency |
| R7 | 4.7 kΩ | Resistor | Based on the desired Frequency |
| R8 | 4.7 kΩ | Resistor | Based on the desired Frequency |
| R9 | 30 Ω | Resistor | LED Current Limiting |
| C3 | 100 nF | Capacitor | Timing or Decoupling |
| C2 | 100 nF | Capacitor | Timing |
| C10 | 10 nF | Capacitor | Timing |
| C6 | 10 nF | Capacitor | Timing |
| IR Emitter | IR LED | Diode | Infrared output |

## Receiver Components

| Reference | Value | Component Type | Stage / Notes |
|---|---|---|---|
| D5 | Photodiode | Diode | First Stage Amplifier |
| R20 | 100 kΩ | Resistor | First Stage Amplifier |
| R21 | 1 kΩ | Resistor | First Stage Amplifier |
| R22 | 2.2 kΩ | Resistor | First Stage Amplifier |
| OA1 | LM660 | Op-Amp | First Stage Amplifier |
| R30 | 470 kΩ | Resistor | Signal Filtering |
| R31 | 1.2 kΩ | Resistor | Signal Filtering |
| R32 | 47 kΩ | Resistor | Signal Filtering |
| C20 | 6.8 nF | Capacitor | Signal Filtering |
| C26 | 6.8 nF | Capacitor | Signal Filtering |
| OA2 | LM660 | Op-Amp | Signal Filtering |
| R33 | 470 kΩ | Resistor | Signal Filtering |
| R34 | 1.2 kΩ | Resistor | Signal Filtering |
| R35 | 47 kΩ | Resistor | Signal Filtering |
| C27 | 680 pF | Capacitor | Signal Filtering |
| C28 | 680 pF | Capacitor | Signal Filtering |
| OA12 | LM660 | Op-Amp | Signal Filtering |
| R36 | 1.5 kΩ | Resistor | Amplification |
| R37 | 100 kΩ | Resistor | Amplification |
| R38 | 1 kΩ | Resistor | Amplification |
| R39 | 1 MΩ | Resistor | Amplification |
| R40 | 1 kΩ | Resistor | Amplification |
| R41 | 2.2 kΩ | Resistor | Amplification |
| R42 | 2.2 kΩ | Resistor | Amplification |
| C29 | 10 nF | Capacitor | Amplification |
| C30 | 100 nF | Capacitor | Amplification |
| C31 | 100 nF | Capacitor | Amplification |
| OA13 | LM660 | Op-Amp | Amplification |
| OA14 | LM660 | Op-Amp | Amplification |

| Reference | Value | Component Type | Stage / Notes |
|---|---|---|---|
| D6 | 700 mV | Diode | Output |
| D10 | 700 mV | Diode | Output |
| R23 | 1 kΩ | Resistor | Output |
| R25 | 100 kΩ | Resistor | Output |
| R26 | 1 kΩ | Resistor | Output |
| R43 | 1 kΩ | Resistor | Output |
| R44 | 100 kΩ | Resistor | Output |
| R45 | 1 kΩ | Resistor | Output |
| R46 | 1 kΩ | Resistor | Output |
| R47 | 1 kΩ | Resistor | Output |
| C23 | 1 µF | Capacitor | Output |
| C32 | 100 mF | Capacitor | Output |
| OA9 | LM660 | Op-Amp | Output |
| OA15 | LM660 | Op-Amp | Output |

# Reference

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